

From Fragmented Records to Actionable Intelligence: LACK, a Knowledge Graph for Climate Accountability Practitioners

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Abstract. Climate accountability work, carried out by investigative journalists, NGOs, and watchdog organisations, depends on knowing who lobbies against climate action and how these actors are connected. This evidence is documented in depth by platforms such as DeSmog and LobbyMap, but remains fragmented, heterogeneous, and hard to analyse at scale. We present LACK (Lobbying Agents, Climate, and Knowledge), to our knowledge the first openly published knowledge graph of climate lobbying networks: ~28k entities and ~47k asserted relations (~189k after inference), built from over 2k web pages. By making these relationships queryable, LACK is designed to support the tasks practitioners perform: tracing funding flows, tracking how actors’ positions shift over time, and mapping influence networks across organisations. We demonstrate LACK through case studies, report how domain experts assess its value, and discuss the opportunities and challenges of building such resources automatically in this domain.

1 Introduction

Lobbying networks that undermine public trust in climate science and delay effective policy action are extensively documented by investigative journalists and civil society organisations, including dedicated platforms such as DeSmog [17] and InfluenceMap’s LobbyMap [22], which together publish nearly 2000 profiles of people and organisations who lobby against climate action and policy. Despite the breadth of such work, the resulting information remains siloed across heterogeneous platforms and is published primarily in long-form prose, organised around organisations’ and individuals’ profiles. As a consequence, even basic questions, such as who funds a given think tank, which staff previously worked for fossil fuel companies, or which board members sit across multiple front groups, require tedious manual cross-referencing across sites with incompatible structures. Investigative journalists, NGOs, and watchdog organisations rely on this

kind of manual analysis, which does not scale. Representing this domain as a knowledge graph makes such relational knowledge explicit and queryable at scale.

In this paper, we present LACK (Lobbying Agents, Climate, and Knowledge), a knowledge graph of persons and collectives involved in lobbying against climate science and policy, built from over 2k source documents and guided by a purpose-built lightweight ontology. We make three contributions. First, we present the LACK resource: a purpose-built vocabulary covering the key relation types in the climate lobbying domain, and a knowledge graph of typed relations between persons and organisations, extracted from authoritative sources, linked to Wikidata and DBpedia, and published as Linked Open Data. Second, we demonstrate the analytical value of LACK through case studies that surface connections spanning multiple independent sources, and we assess the resource both for its coverage of the competency questions behind its design and through a survey of domain experts in investigative journalism and climate accountability. Third, drawing on both the construction process and this expert engagement, we contribute an analysis of the opportunities and open challenges of building such resources automatically in this domain, including temporal change, funding traceability, shifting positions, long-tail entities, and geographic skew.

The remainder of this paper is structured as follows. Section 2 situates LACK in the related literature. Section 3 describes how LACK is built and what it contains, covering the ontology, the extraction process, and the resulting resource. Section 4 reports the case studies and the expert evaluation. Section 5 discusses the opportunities and open challenges of building and using such a resource in practice, and concludes with directions for future work.

2 Background and Related Work

Corporate lobbying has demonstrable consequences for climate policy outcomes. Meng and Rode [25] estimate that lobbying lowered the probability of enacting the Waxman–Markey Cap-and-Trade Bill by 13 percentage points, with an estimated welfare loss of roughly \$60 billion, while Baehr, Bare and Heddesheimer [4] show across more than 11,000 firms that earnings-call *climate exposure* systematically predicts whether, how much, and where a firm lobbies. Understanding lobbying therefore requires more than aggregate spending figures: it requires modelling the underlying network of actors, funders, and affiliations through which climate-policy influence is organised and exerted.

The resources practitioners rely on today partly address the documentation challenge but stop short of making the information and network integrated and queryable. DeSmog [17] and LobbyMap [22] profile, respectively, denial actors and corporate policy engagement, Skeptical Science [33] curates attributed statements, and Oreskes and Conway [28] analyse doubt-manufacturing strategies. These resources are invaluable but heterogeneous, siloed, and not interlinked in any form amenable to computational analysis, leaving cross-source questions to manual effort.

Closest to our work is the UCL *AI-Powered Climate Lobbying Database* [23], a concurrent project that likewise proposes to apply large language models to extract lobbying information from text and construct network graphs. Its stated

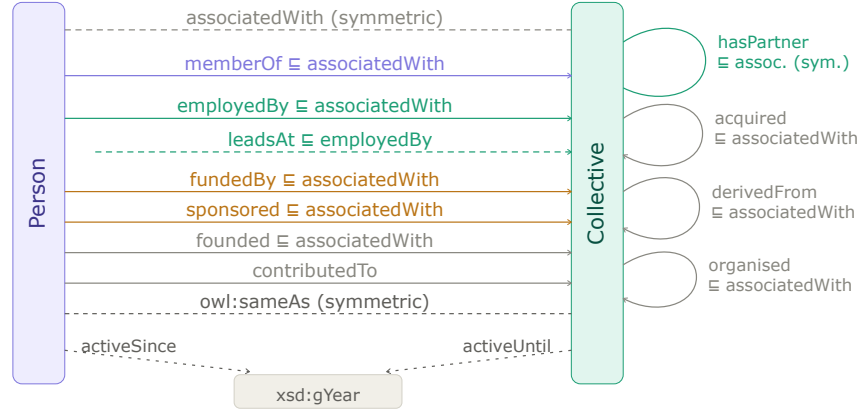


Fig. 1. Schema diagram of the LACK ontology.

emphasis is on policy-position analysis, whereas LACK models the actors themselves and the affiliation, funding, and personnel ties between them. We are not aware of a public release of the resulting data at the time of writing.

In the Semantic Web community, CimpleKG [8] provides a continuously updated knowledge graph on misinformation and fact-checks across domains, currently being extended within the ClimateSense project [10] with climate-specific classifiers, while ClimaFactsKG [7] structures scientific evidence countering climate misinformation; both, however, target claim verification rather than lobbying actor networks. Knowledge graphs have also been adopted as infrastructure for investigative journalism [27,19,6,1,26] and for political discourse, including European Parliament debates [35] and the digitised Canadian parliamentary record [5], but these model news content, events, or formal proceedings rather than the off-stage networks of funding, affiliation, and personnel through which lobbying influence operates. Constructing such a graph relies on well-studied information extraction and entity resolution and linking techniques [9,32,3], increasingly addressed with large language models that match or exceed supervised baselines without task-specific training [29], in combination with large open knowledge bases such as Wikidata. It is a question how those fit the climate-lobbying setting, where many mentions are long-tail entities absent from general-purpose knowledge bases [20].

3 The LACK Resource

Sources. LACK was built from the two most comprehensive English-language databases dedicated to climate-lobbying actors⁴. The **DeSmog Climate Disinformation Database** [17], maintained by DeSmog (an international investigative journalism organisation founded in 2006 and philanthropically funded), profiles individuals and organisations involved in climate denial, typically covering

⁴ Pages accessed in December 2025.

positions on climate science, affiliations, funding sources, and public statements; we collected all 943 profile pages. **LobbyMap** [22], maintained by InfluenceMap (an independent non-profit think tank founded in 2015), provides analyst-scored, IPCC-benchmarked assessments of corporate and industry-association engagement with climate policy for approximately 1,000 companies and 330 associations; we collected the 964 LobbyMap Scores profiles plus 338 further organisations referenced within them (1,302 pages total). The two sources are complementary: DeSmog focuses on individuals and think tanks involved in disinformation, while LobbyMap covers groups engaging with policy; well-known actors appear in both. The pages were collected using SPARQL Anything [16], querying DeSmog’s HTML pages’ index and the LobbyMap leaderboard JSON feed.

Ontology. The LACK ontology (Figure 1) was designed following the NeOn methodology [34], with one deliberate adaptation: rather than eliciting requirements from domain experts directly, we treated the investigative sources themselves as a proxy for user needs, assuming that platforms such as DeSmog and LobbyMap already encode the questions their users care about. To guide design, we derived twelve *competency questions* (following [34]) from information content of the source pages, and re-engineered those sources, as non-ontological resources, into a domain schema⁵; we revisit and confirm this assumption through the expert evaluation in Section 4. Practitioner requirements were not elicited at design time, so the schema is bounded by what the sources themselves document; the expert survey (Section 4.3) provides post-hoc validation and, as we discuss there, marks that boundary. Crucially, relation types that could not be derived from the sources are excluded. We evaluated general-purpose vocabularies such as Schema.org [30] or the W3C Organisation ontology, which don’t have lobbying-specific relations and/or would fragment the graph through intermediary role nodes. The ontology defines two top-level classes, `lack:Person` (an individual) and `lack:Collective` (any group organised around a shared purpose, deliberately broad to accommodate events, journals, campaigns, and groups alongside organisations)⁶. Each relation is defined with a domain, range, and inverse, organised in a hierarchy rooted at the symmetric catch-all `lack:associatedWith`. Relationships can be time-bounded, captured through entity-level bounds (`lack:activeSince`/`lack:activeUntil`) and relation-level annotations via reification [31], where each temporally annotated triple is reified as an `rdf:Statement` carrying `lack:since`/`lack:until`; reification was chosen over RDF-star for broad compatibility with current RDF tooling. Provenance and source attribution reuse PROV-O [24] and Dublin Core Terms [18], linking each statement to the extraction activity and to the source web page.

Construction. The knowledge graph is produced from the raw extraction outputs by a pipeline implemented with SPARQL Anything [2], reproducible from

⁵ <https://climatesense-project.eu/lack/ontology.html>

⁶ We note that alignments with `foaf:Agent` and `foaf:Person` are possible for `lack:Person` but not for `lack:Collective`.

Table 1. LACK KG statistics. Inferred triples overlap with asserted ones, so the post-inference count is not their sum. The graph total additionally includes reified statements, provenance, and `owl:sameAs/rdfs:seeAlso` links.

Dimension	Count
Entities (Person / Collective / Total)	10,370 / 17,604 / 27,945
Wikidata links (Person / Collective / Total)	2,911 / 5,369 / 8,280
DBpedia links (Person / Collective / Total)	3,516 / 5,210 / 8,726
Relations triples, asserted	47,450
Relations triples, after inference	188,862
All triples in the graph	213,147
Evidence (Desmog / LobbyMap / InfluenceMap / Total)	943 / 964 / 338 / 2,245

the source CSVs [14]. Relations are first extracted from the source pages by prompting Gemini 2.5 Flash [11] with the allowed relation types and a strict JSON schema, then linked to authoritative identifiers through a hybrid entity-linking step combining Wikidata candidate retrieval, LLM-based disambiguation, and a web-search fallback, with Llama 3.3 70B [21] used at scale [15,13]. Construction then proceeds in five phases: (i) the 152 distinct surface-form relation types (90,145 instances) are aligned to ontology properties; (ii) stable IRIs are minted per entity by hashing the highest-priority identifier available (Wikidata QID, then Wikipedia, then web page, then surface form with source URL), each typed as `lack:Person` or `lack:Collective` with a granular subtype and, where available, `owl:sameAs` links to Wikidata and DBpedia; (iii) each relation is written as a typed triple plus a reified statement carrying evidence and temporal bounds; (iv) PROV-O provenance is attached; and (v) ontology entailments (inverse pairs, sub-property promotions, symmetric closure) are materialised, growing the graph from 47,450 asserted relations to 188,862 triples.

Availability and Maintenance. The LACK ontology and knowledge graph are published as Linked Open Data under a Creative Commons Attribution Non-Commercial 4.0 International (CC BY NC 4.0) licence. Version 1.1 was released in April 2026 and is accessible at <https://climatesense-project.eu/lack/> as a Turtle dump (KG.ttl), a live SPARQL endpoint powered by QLever⁷, and an interactive exploration interface. Table 1 summarises statistics; a full breakdown is on the LACK website. The ontology is separately available under the persistent namespace <https://purl.net/climatesense/lack/ns#>, and the dataset is archived on Zenodo [12]. The resource is maintained by KMi and guaranteed available for at least five years under funder open data policy. Infrastructure for continuous updates is currently under development.

4 Demonstration and Evaluation

4.1 Resource Quality and Coverage

We evaluated relation extraction on a sample of 300 relations drawn from 8 randomly selected DeSmog profile pages, with three annotators working in pairs

⁷ <https://sparql.climatesense.kmi.tools/climatesense>

and disagreements resolved by discussion, assessing each relation for truthfulness, validity, and temporal correctness⁸. Truthfulness accuracy is 0.93 (0.99 with partial credit), validity 0.91 (0.95 with partial), and temporal correctness 0.80 (0.91 with partial), where partial credit captures approximately correct extractions; temporal correctness is the weakest dimension, with errors from missing dates and ambiguity between point-in-time and interval information. Entity linking, evaluated on 225 entities against a curated gold standard⁹, is accurate across the primary output fields (Wikidata URI and Wikipedia URL both 0.88, parent entity 0.94). Of the 27,945 entities, 36.5% are linked to Wikidata or DBpedia; the high NIL rate is a domain characteristic rather than a pipeline failure, since the climate-lobbying ecosystem is densely populated with minor think-tank staff, obscure consultants, short-lived coalitions, and regional trade bodies absent from general-purpose knowledge bases. Tellingly, entities documented by both sources, those prominent enough to appear independently in DeSmog and LobbyMap, are linked at 95.7% (488 of 510), confirming that the pipeline reliably grounds well-known actors.

We also assess how well the graph answers the twelve competency questions that drove its design, classifying each into a coverage tier by the share of entities it reaches. Coverage is solid across the core transparency and network-structure questions: employment, membership, leadership, and statement-level temporal annotation all reach the Strong tier, while funding and structural connections are Moderate and founding relations remain Weak, reflecting the difficulty of reliably extracting founding events. The most striking result is network reach: after inferencing, 93.5% of entities are reachable via `lack:associatedWith`, turning a sparse set of assertions into a near-complete connectivity structure. Temporal coverage is partial, with statement-level annotation reaching the Strong tier but entity-level activity bounds covering only 2.6% of entities.

4.2 Case Studies

Integrating the sources. LACK’s central value is making visible connections that no single source exposes. For example, InfluenceMap records Darren Woods leading Exxon (`lack:leadsAt`) while DeSmog records Exxon sponsoring the Heartland Institute (`lack:sponsored`), a think tank associated with climate science denial; joining the two relations in a single SPARQL query links a named corporate officer to a denial organisation through his employer. In another example, DeSmog records a partnership between Cambridge Analytica and BP (`lack:hasPartner`) while InfluenceMap records BP’s membership of the Business Council of Australia (`lack:memberOf`); the two facts originate in different investigative contexts and share only BP, yet LACK surfaces a cross-domain, cross-jurisdictional connection invisible to either source alone. LACK includes 27,757 of such cross-source two-hops paths.

Cross-referencing with COP30. Pairing LACK with an external source of named climate-policy agents enables further analyses. We cross-referenced LACK’s

⁸ Full annotation guidelines: https://bit.ly/lack_42bhqzl.

⁹ Gold standard preparation: https://bit.ly/lack_3QQPRsD.

1,000 most heavily-described `lack:Collective` entities with the COP30 participant list via heuristic label matching. Of 944 distinct delegate matches, roughly 70% flow through observer accreditation channels, 18% appear in high-income national delegations, and only 43 in lower- or low-income country delegations. Within the most densely connected entities, the US climate countermovement (Heartland, Cato, ALEC, Heritage, Manhattan Institute) is almost absent from COP30, with only three (US Chamber of Commerce, Shell, ExxonMobil) present at all. The cross-reference surfaces two findings invisible to either resource alone: the US climate countermovement does not engage with the UNFCCC track, and observer accreditation, rather than national delegation, is the dominant channel through which LACK-tracked collectives participate in COP.

4.3 Expert Evaluation

A 14-question online questionnaire was distributed via purposive sampling to researchers, fact-checkers, investigative journalists, NGO professionals, and policy analysts working on climate lobbying networks, across 25 organisational groups spanning academic networks, fact-checking organisations, investigative outlets, NGOs, and international bodies. Nine domain experts were involved in the evaluation: three researchers, alongside a fact-checker, a policy analyst, an educator, and an NGO professional, with one respondent spanning researcher, journalist, and NGO roles. Five of them work with climate lobbying information daily or weekly, so these are active practitioners, and while the sample is small, it spans the primary user communities, with responses consistent across professional backgrounds. Respondents unanimously rated both the overall value of such a resource and the ability to search across multiple sources at 4.62 on a 5-point scale, directly validating LACK’s integration approach. On priority relation types, funding flows received the highest mean rating (4.75), followed by employment (4.50), and membership, sponsorship, and founding (each 4.38). These priorities track our coverage closely: employment and membership are both well served, whereas funding, the top-rated category, is only moderately covered, consistent with respondents’ own remarks on how hard funding information is to obtain. Temporal information was rated $M=3.88$ and flagged by half the respondents as one of the hardest dimensions to find, consistent with the partial temporal coverage reported above. Several respondents identified gaps outside LACK’s current scope, notably direct connections to the political sphere and a need to distinguish intermediary actor types (lobby consultancies, law firms, PR agencies), and recommended integrating further sources such as the EU Transparency Register, LobbyFacts, SourceWatch, and OpenSecrets. We return to these gaps, and to the broader challenges they raise, in Section 5.

5 Discussion and Outlook

Building LACK showed that valuable info can be automatically extracted, integrated and enriched with semantics. Connections that are invisible within any single source become queryable once the sources are represented in a common graph, and this cross-source bridging is precisely what practitioners told us they

value most. We see no reason this approach should remain specific to climate lobbying.

The same pattern, harvesting heterogeneous investigative documentation, extracting typed relations with large language models, and grounding entities against open knowledge bases, could support accountability work in other domains where influence is exercised through networks of funding, affiliation, and personnel, from public-health lobbying to defence procurement. Demonstrating this generalisation is an aim of our ongoing work.

At the same time, the construction process surfaces challenges that are characteristic of this domain. Lobbying networks are dynamic: positions, affiliations, and roles change over time, yet temporal information is the dimension our evaluation and our experts both found hardest to capture, and entity-level activity bounds remain sparse. Funding is the relation practitioners most want and one of the hardest to extract reliably, since funding ties are often disclosed obliquely or not at all, as our experts also noted. The long tail of minor actors, obscure consultants, short-lived coalitions, regional trade bodies, drives the high proportion of entities without external identifiers; rather than a mere limitation, this is an open research problem, since enriching and disambiguating long-tail entities absent from general-purpose knowledge bases is an active challenge for automatic knowledge graph construction [20]. The reliance on English-language sources introduces a geographic and linguistic skew, visible in our COP30 cross-reference, where actors from lower-income countries are markedly under-represented.

Turning a resource into something practitioners can adopt raises questions beyond extraction quality. LACK reorganises information produced by investigative organisations who invest considerable effort in it, and respecting that effort is of primary importance: we followed DeSmog’s republishing guidelines and complied with the redistribution terms we obtained from InfluenceMap, and we publish LACK under a Creative Commons Attribution-NonCommercial licence. In an era when large commercial actors routinely ingest others’ content without credit, a BY-NC licence preserves attribution and reserves commercial rights to the original data owners, which we hope makes it easier, not harder, for further investigative organisations to contribute to a shared LACK data space. The gaps our experts identified, direct links to the political sphere, finer distinctions between intermediary actor types such as consultancies and PR agencies, and integration of further sources such as the EU Transparency Register and OpenSecrets, define our agenda for that growth.

In summary, LACK turns fragmented investigative records on climate lobbying into a queryable resource whose quality and reception we have assessed with the practitioners it is meant to serve. By releasing it as an open, attribution-preserving resource, we hope to invite the organisations whose work it builds on to help extend it, growing it from a knowledge graph into a shared data space for climate accountability.

References

1. Alexander, R.J., Bartocci, M., Persico, O., Vetere, G.: Harnessing ‘Il Manifesto’ newspaper archive for knowledge base creation: Techniques and findings in the

- ‘MeMa’ project. In: Proceedings of the 9th Italian Conference on Computational Linguistics (CLiC-it 2023). pp. 28–35. CEUR Workshop Proceedings (2023), <https://aclanthology.org/2023.clicit-1.5/>
2. Asprino, L., Daga, E., Gangemi, A., Mulholland, P.: Knowledge graph construction with a façade: A unified method to access heterogeneous data sources on the web. *ACM Transactions on Internet Technology* **23**(1), 1–31 (2023)
 3. Ayoola, T., Tyagi, S., Fisher, J., Christodoulopoulos, C., Pierleoni, A.: ReFinED: An efficient zero-shot-capable approach to end-to-end entity linking. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Industry Track. pp. 209–220. Association for Computational Linguistics (2022), <https://aclanthology.org/2022.naacl-industry.24/>
 4. Baehr, C., Bare, F., Heddeshheimer, V.: Climate exposure drives firm political behavior: Evidence from earnings calls and lobbying data. *American Journal of Political Science* (2026). <https://doi.org/10.1111/ajps.70062>, early View
 5. Beelen, K., Thijm, T.A., Cochrane, C., Halvemaan, K., Hirst, G., Kimmins, M., Lijbrink, S., Marx, M., Naderi, N., Rheault, L., et al.: Digitization of the canadian parliamentary debates. *Canadian Journal of Political Science/Revue canadienne de science politique* **50**(3), 849–864 (2017)
 6. Berven, A., Christensen, O.A., Moldeklev, S., Opdahl, A.L., Villanger, K.J.: A knowledge-graph platform for newsrooms. *Computers in Industry* **123**, 103321 (2020). <https://doi.org/10.1016/j.compind.2020.103321>
 7. Burel, G., Alani, H.: Climafactskg: Towards an interlinked knowledge graph of scientific evidence to fight climate misinformation. In: CEUR Workshop Proceedings. vol. 4065, pp. 134–140 (2025)
 8. Burel, G., Mensio, M., Peskine, Y., Troncy, R., Papotti, P., Alani, H.: Cimplekg: A continuously updated knowledge graph on misinformation, factors and fact-checks. In: International Semantic Web Conference. pp. 97–114. Springer (2024)
 9. Christophides, V., Efthymiou, V., Palpanas, T., Papadakis, G., Stefanidis, K.: An overview of end-to-end entity resolution for big data. *ACM Computing Surveys* **53**(6), 127:1–127:42 (2020). <https://doi.org/10.1145/3418896>
 10. ClimateSense Project: ClimateSense: Climate misinformation surveillance and analysis with multidimensional GIS. <https://climatesense-project.eu/> (2025), accessed: 2026-04-09
 11. Comanici, G., Bieber, E., Schaekermann, M., et al.: Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities (2025), <https://arxiv.org/abs/2507.06261>
 12. Daga, E.: enridaga/lack-zenodo-deposit: v1.1 (Apr 2026). <https://doi.org/10.5281/zenodo.19847768>, <https://doi.org/10.5281/zenodo.19847768>
 13. Daga, E.: Lack entity linking (enridaga/lack-el: v1.0) (2026). <https://doi.org/10.5281/zenodo.20035041>
 14. Daga, E.: LACK Knowledge Graph Construction (2026). <https://doi.org/10.5281/zenodo.19861384>
 15. Daga, E.: Lack relation extraction notebooks — supplementary material (2026). <https://doi.org/10.5281/zenodo.20036778>
 16. Daga, E., Asprino, L., Mulholland, P., Gangemi, A.: An opinionated approach to sparql anything. In: Further with Knowledge Graphs: Proceedings of the 17th International Conference on Semantic Systems, 6-9 September 2021, Amsterdam, The Netherlands. vol. 53, p. 58. SAGE Publications Limited (2021)
 17. DeSmog: Climate disinformation database. <https://www.desmog.com/climate-disinformation-database/> (2025), accessed: December 2025

18. Dublin Core Metadata Initiative: DCMI metadata terms. Dcmi recommendation, Dublin Core Metadata Initiative (2020), <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/>
19. Gallofré Ocaña, M., Opdahl, A.L.: Supporting newsrooms with journalistic knowledge graph platforms: Current state and future directions. *Technologies* **10**(3), 68 (2022). <https://doi.org/10.3390/technologies10030068>
20. Graciotti, A., Lazzari, N., Daga, E., Presutti, V.: Lost and found: Enriching knowledge graphs with “nil” persons from historical documents. *Semantic Web* (2026), to appear
21. Grattafiori, A., et al.: The Llama 3 herd of models. arXiv preprint arXiv:2407.21783 (2024), <https://arxiv.org/abs/2407.21783>
22. InfluenceMap: LobbyMap Scores. <https://lobbymap.org/LobbyMapScores> (2025), accessed: December 2025
23. Jordan, N., Coen, D.: AI-Powered Climate Lobbying Database: Automating text extraction, network graphs, and policy position analysis. UCL Grand Challenges, Climate Crisis Pump Priming Grant (2025), <https://www.ucl.ac.uk/grand-challenges/case-studies/2025/jun/ai-powered-climate-lobbying-database>, accessed April 2026
24. Lebo, T., Sahoo, S., McGuinness, D., et al.: PROV-O: The PROV ontology. W3c recommendation, W3C (2013), <https://www.w3.org/TR/prov-o/>
25. Meng, K.C., Rode, A.: The social cost of lobbying over climate policy. *Nature Climate Change* **9**(6), 472–476 (2019). <https://doi.org/10.1038/s41558-019-0489-6>
26. Motta, E., Daga, E., Gangemi, A., Gjelsvik, M.L., Osborne, F., Salatino, A.: The epistemology of fine-grained news classification. *Semantic Web* **16**(3), 1–26 (2025). <https://doi.org/10.1177/22104968251344461>
27. Opdahl, A.L., Al-Moslmi, T., Dang-Nguyen, D.T., Gallofré Ocaña, M., Tessem, B., Veres, C.: Semantic knowledge graphs for the news: A review. *ACM Computing Surveys* **55**(7), 1–38 (2022)
28. Oreskes, N., Conway, E.M.: *Merchants of doubt: How a handful of scientists obscured the truth on issues from tobacco smoke to global warming*. Bloomsbury Publishing USA (2011)
29. Peeters, R., Steiner, A., Bizer, C.: Entity matching using large language models. In: *Proceedings of the 28th International Conference on Extending Database Technology (EDBT)* (2025). <https://doi.org/10.48786/edbt.2025.42>
30. Schema.org Community Group: Schema.org. <https://schema.org/> (2011)
31. Schreiber, G., Raimond, Y.: RDF 1.1 primer. W3c note, W3C (2014), <https://www.w3.org/TR/rdf11-primer/>
32. Sevgili, Ö., Shelmanov, A., Arkhipov, M., Panchenko, A., Biemann, C.: Neural entity linking: A survey of models based on deep learning. *Semantic Web* **13**(3), 527–570 (2022). <https://doi.org/10.3233/SW-222986>
33. Skeptical Science: Skeptical science: Climate misinformation by source. <https://skepticalscience.com/skepticquotes.php> (2025), accessed: April 2026
34. Suárez-Figueroa, M.C., Gómez-Pérez, A., Fernández-López, M.: The neon methodology for ontology engineering. In: *Ontology engineering in a networked world*, pp. 9–34. Springer (2011)
35. Van Aggelen, A., Hollink, L., Kemman, M., Kleppe, M., Beunders, H.: The debates of the european parliament as linked open data. *Semantic Web* **8**(2), 271–281 (2016)